

Alexandre Roy

Curriculum Vitae

Education

- 2024–Present **PhD student in Mathematics**, *Université de Caen Normandie*, Caen, Advisor : Jérôme Poineau.
Title : Compactification hybride d'espaces de modules de systèmes dynamiques
- 2023 **Master's Degree in Pure Mathematics**, *Université de la Sorbonne*, Paris.
Algebraic geometry and number theory
- 2021 **Bachelor's Degree in Mathematics**, *Université de la Sorbonne*, Paris.
- 2020–2024 **Diplôme de l'ENS**, *Ecole Normale Supérieure - Ulm*, Paris.
Département de Mathématiques
- 2018–2020 **Classe Préparatoire aux Grandes Ecoles**, *Lycée Henri IV*, Paris.
- 2018 **Bccalauréat Scientifique**, *Lycée Hector Berlioz*, Vincennes.

Experience

- 2023–2024 **Research Internship**, *Université de Caen Normandie*, Caen, Advisor : Jérôme Poineau.
Some readings about the future PhD thesis
- Mar. - Jun. 2023 **Research Intership, Master's thesis**, *Universität Johann Wolfgang Goethe*, Frankfurt, Advisor : Jérôme Poineau.
Title : p -adic and arithmetic dynamics via Berkovich spaces
- Mar. - Jun. 2022 **Research Intership**, *University of Copenhagen*, Copenhagen, Advisors : Jasmin Matz and Morten Riesager.
Title : Computing number fields with fixed degree and bounded discriminant
- May - Jun. 2021 **Mémoire de 1ère année**, with Jack Souami, *Ecole Normale Supérieure*, Paris, Advisor : Arnaud Eteve.
Title : Construction of the exponential function in positive characteristic

Organization

- Dec. 4–6 2025 **Number Theory Team's Retreat**, *Université de Caen Normandie*, Bayeux, with Denis Simon.
Mini conference between the members of the Number Theory's Team
- 2025–Present **Council member of the LMNO**, *Université de Caen Normandie*, Caen.
Representative of the PhD students
- May 24–26 2025 **PhD student's Retreat**, *Université de Caen Normandie*, Bayeux.
Mini conference between the PhD students of Caen

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- Nov. 28-30 **Number Theory Team's Retreat**, *Université de Caen Normandie*, Bayeux, with
 2024 Alexis Lucas.
 Mini conference between the members of the Number Theory's Team

Publications

- 2025 **Action de groupe algébrique sur la compactification hybride**, *preprint*.

Teaching

- 2025-2026 **Algebra**, *first year mathematics students*, Université de Caen Normandie, linear systems, polynomials, real vector spaces.
 50h of mixed courses and exercises sessions + 4h of preparation for the examination
General Mathematics, *first year economic and legal students*, Université de Caen Normandie, 3x3 matrices, graphs.
 10h of exercises sessions
- 2024-2025 **Algebra**, *first year mathematics students*, Université de Caen Normandie, linear systems, polynomials, real vector spaces.
 50h of mixed courses and exercises sessions
General Mathematics, *first year economic and legal students*, Université de Caen Normandie, 3x3 matrices, graphs.
 10h of exercises sessions
- 2020-2022 **Mock Oral Exams (colle)**, *first year CPGE students*, Lycée Henri IV, Algebra, analysis and probability.
 32h each year

Talk

- 02/07/2026 **On the hybrid compactification of M_d** , *Conference : New trends in holomorphic dynamics in dimension one and higher*, (Orléans, France).
- 29/05/2026 **Sur la compactification hybride de M_d** , *Monthly Complex dynamics Seminar*, LPSM (Paris, France).
- 30/04/2025 **Sur la compactification de M_d** , *Young Reasearcher Seminar : Very short talks*, LMNO (Caen, France).
- 16/04/2026 **Introduction à la dynamique non-archimédienne II**, *Young Reasearcher Seminar*, LMNO (Caen, France).
- Nov. 27 2025 **Introduction à la dynamique non-archimédienne**, *Young Reasearcher Seminar*, LMNO (Caen, France).
- Nov. 10 2025 **Introduction à la mesure de Mahler**, *Working group about Lehmer's problem*, LMNO (Caen, France).
- Jun. 12 2025 **Sur la continuité des ensembles de Julia**, *Young Reasearcher Seminar : Very short talks*, LMNO (Caen, France).
- Mar. 21 2025 **Definitions of ∞ -categories II**, *Working group about ∞ -categories*, LMNO (Caen, France).

- Mar. 7 2025 **Definitions of ∞ -categories I**, *Working group about ∞ -categories*, LMNO (Caen, France).
- Nov. 28 2024 **Sur l'article de R. Rumely The minimal resultant locus**, *Number Theory Team Retreat*, (Bayeux, France).
- Jun. 20 2024 **Espaces de Berkovich et dynamique**, *Young Researcher Seminar*, LMNO (Caen, France).
- Jun. 27 2022 **The number of number fields with fixed degree and bounded discriminant**, *Number Theory Seminar*, University of Copenhagen (Copenhagen, Denmark).

Training

- 2026-Present **Working group**, *p-adic Operator's Algebra*, Université de Caen Normandie, Caen (France).
- 2026-Present **Working group**, *Light condensed mathematics*, Université de Caen Normandie, Caen (France).
- 18/05/2026 – 22/05/2026 **ANR meeting**, *Adelic, Analytic, Arithmetic*, CMLS, Palaiseau (France).
- 2025-Present **Working Group**, *Autour du problème de Lehmer*, Université de Caen Normandie, Caen (France).
- 03/09/2025 – 05/09/2025 **ANR meeting**, *Arithmetic, Algebraic and Analytics Dynamics*, Université de Bourgogne, Dijon (France).
- 2025-2026 **Working Group**, *Berkovich Motives*, Université de Caen Normandie, Caen (France).
- Jan. 27-31 2025 **Workshop**, *Model-theoretic methods for non-archimedean geometry*, University of Münster, Münster (Germany).
- Jan. 20-24 2025 **Research School**, *Arithmetic, Algebraic and Analytics Dynamics*, CIRM, Marseille (France).
- 2024-2025 **Working Group**, *∞ -categories*, Young researchers working group, Université de Caen Normandie, Caen (France).
- Mar. 4-8 2024 **Spring School**, *Non-archimedean Geometry*, Heildeberg Universität, Heidelberg (Germany).

Languages

French	Native	
English	Proficient user	<i>Con conversationally fluent</i>
German	Basic/independent user	

Programming

Latex	proficient user
Python	independent user
HTLM	beginner user